

Ideation Phase

Empathize & Discover

Date	13 July 2026
Team ID	[To be filled by student/team]
Project Name	Voice-Based Concept Understanding Analyzer (VBCUA)
Maximum Marks	4 Marks

Empathy Map Canvas

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes. It is a useful tool that helps teams better understand their users.

Creating an effective solution requires understanding the true problem and the person experiencing it. The exercise of creating this map helped the team consider the Voice-Based Concept Understanding Analyzer (VBCUA) from the user's perspective, along with their goals and challenges.

Reference: <https://www.mural.co/templates/empathy-map-canvas>

Example: Voice-Based Concept Understanding Analyzer (VBCUA)

PERSONA

The Student — Primary User of VBCUA

A learner who records a spoken explanation of an academic concept and wants to know, quickly and objectively, how well that explanation demonstrates real understanding.

THINK & FEEL

- “Am I actually explaining the concept correctly, or just repeating memorised phrases?”
- Worried about losing marks for missing key ideas without realising it.
- Hopes for fast, objective feedback instead of waiting days for a grade.
- Cares about genuine conceptual understanding, not rote memorisation.

HEAR

- Instructors repeatedly stressing “explain it in your own words.”
- Classmates comparing how detailed or complete their spoken answers were.
- Feedback on earlier assignments noting “missing key points.”
- General discussion about AI tools that can evaluate spoken answers.

SEE

- A simple upload screen for submitting a recorded audio explanation.
- Other students receiving instant percentage-based scores on written quizzes.
- The reference answer / marking rubric shared by the instructor.
- Peers frustrated by slow, manual grading turnaround.

SAY & DO

- Records or uploads a spoken explanation of the assigned concept.
- Mentally rehearses the answer before recording it.
- Checks that the audio file matches a supported format (WAV, MP3, M4A).
- Reviews the downloaded PDF report and written feedback afterward.
- Re-records or revises understanding when the score is low.

PAIN

- Fear of being graded inconsistently or unfairly compared with classmates.
- No way to check personal understanding before final submission.
- Uncertainty about exactly which concepts were missed or misunderstood.
- Long turnaround time to receive feedback from a human instructor.

GAIN

- An immediate, quantifiable concept-understanding score to gauge readiness.
- Clear feedback indicating which concept areas were strong or weak.
- A downloadable report that documents progress over time.
- The ability to self-correct and practise again before formal evaluation.